UNIVERSITI MALAYA

EXAMINATION FOR THE DEGREE OF MASTER OF SOFTWARE ENGINEERING

ACADEMIC SESSION 2025/2026: SEMESTER II

WOC7017: BIG DATA PROCESSING

MAY-JUNE, 2026 WEEK 9 – WEEK 14

ALTERNATIVE ASSESSMENT (40%):

PROJECT PRESENTATION AND REPORT

INSTRUCTION & ASSESSMENT RUBRIC

(This question paper consists of 1 question on 5 printed pages)

COURSE CODE	WOC7017
COURSE NAME	BIG DATA PROCESSING
SEMESTER	2
SESSION	2024/2025
LECTURER	ASSOC. PROF. DR. MUMTAZ BEGUM MUSTAFA
SUBMISSION	WEEK 9-14
ALTERNATIVE ASSESSMENT	PROJECT PRESENTATION AND REPORT (40%)
COURSE LEARNING OUTCOME	CLO3 - EVALUATE THE SUITABILITY OF DIFFERENT PROCESSING TECHNIQUES FOR BIG DATA PROCESSING.

Instructions for Project Presentation and Report

You are required to work in pair to complete the following:

Part 1: Report (30%)

1. Identify an application domain that requires Big Data Processing and Analytics: e.g., healthcare, insurance, e-commerce, finance, education, etc. Provide a suitable title for your project (example: 1) Data Processing and Analytics for Bike-Sharing Systems, 2) Data Processing and Analytics of Hand, Foot and Mouth Disease (HFMD) in Malaysia.
2. Identify the problems/issues faced in the selected application domain that can be portrayed using suitable big data processing and analytics. **You are required to get approval from your course instructor on the suitability of the selected application domain, the identified problem, and the relevancy of the data processing and analytics.**
3. Identify a suitable large dataset (at least 2 GB) for the selected domain that is relevant for the proposed data processing and analytics. You may refer to the given link for suitable large datasets that can be used to discover interesting relationships/patterns that will be useful for this domain in the context of Big Data.

    - https://www.kaggle.com/datasets
    - https://hadoopilluminated.com/hadoop_illuminated/Public_Bigdata_Sets.html
    - https://hadoopproject.com/datasets-for-big-data-projects/
    - https://research.google/resources/datasets/
    - https://github.com/search?q=datasets&type=repositories
    - https://catalog.data.gov/dataset/

4. With the use of Big Data Frameworks (Hadoop and Spark), which can be used to process the identified big or large dataset, you are required to develop suitable MapReduce algorithms using several functions for performing the relevant data processing/analytics. The students are required to solve the selected problem using two different Big Data Frameworks (Hadoop and Spark).
5. Evaluate and compare the performance of the developed MapReduce algorithms using the Hadoop Framework and the Spark Framework in terms of processing time. Discuss the results.
6. Draw a Data Flow Diagram that reflects the developed MapReduce algorithm(s) used for data processing.
7. Write a report that describes the Big Data Analytics Lifecycle stages (data generation, data aggregation, data preprocessing, data integration, data cleaning, data reduction, and data transformation) in data analytics.
8. Submit the project report via Spectrum by week 14.

Part 2: Presentation (10%)

1. Prepare slides and present your project for 20 minutes, including a demo of the implemented algorithm(s). Submit the slides through Spectrum in week 13/14.

TABLE 1: ASSESSMENT RUBRIC

RUBRIC						
Assessment Criteria	**Mark breakdown**	**Score details**				
		1	**2**	**3**	**4**	**MARK TO BE GIVEN**
PART 1: PROJECT REPORT (30%)						
1. Selection of relevant application domain that requires Big Data Processing.	2%	Non-relevant application domain that requires Big Data Processing.	Somehow relevant application domain that requires Big Data Processing.	Relevant application domain that requires Big Data Processing.	Highly relevant application domain that requires Big Data Processing.	**x/4 * 2 = /2**
2. Suitability of the identified problem and the relevancy of the data processing.	2%	The problem is not clear (what are the big data issues and why big data processing is required) for the selected domain. The proposed data processing is not relevant to the identified problem.	The problem is somehow clear (what are the big data issues and why big data processing is required) for the selected domain. The proposed data processing is somehow relevant to the identified problem.	The problem is clear (what are the big data issues and why big data processing is required) for the selected domain. The proposed data processing is relevant to the identified problem.	The problem is very clear (what are the big data issues and why big data processing is required) for the selected domain. The proposed data processing is highly relevant to the identified problem.	**x/4 * 2 = /2**
3. Selection of a suitable large dataset for the selected domain which is relevant for the proposed big data processing.	2%	The dataset is not suitable for the selected domain and does not reflect the big data issue and data processing. The proposed data processing is not relevant to the selected dataset.	The dataset is somehow suitable for the selected domain and somehow reflects the big data issue and data processing. The proposed data processing is somehow relevant to the selected dataset.	The dataset is suitable for the selected domain and reflects the big data issue and data processing. The proposed data processing is relevant to the selected dataset.	The dataset is highly suitable for the selected domain and reflects the big data issue and data processing. The proposed data processing is highly relevant to the selected dataset.	**x/4 * 2 = /2**
4. Number of MapReduce algorithms/ case studies	7%	1	2	3	4	**x/4 * 7 = /7**
5. Number of techniques/functions/ calculations (wordcount, partition, retrieval, average, etc.) used in each implemented	7%	1	2	3	4	**x/4 * 7 = /7**

MapReduce algorithm / case study						
6. Performance evaluation and comparison of the proposed algorithms using two different big data processing frameworks (Hadoop and Spark). Results discussion.	5%	Performance evaluation and comparison of different processing techniques for big data processing are not adequate.	Performance evaluation and comparison of different processing techniques for big data processing are somehow adequate.	Performance evaluation and comparison of different processing techniques for big data processing are adequate.	Performance evaluation and comparison of different processing techniques for big data processing are highly adequate.	**x/4 * 5 = /5**
7. Draw a Data Flow Diagram that reflects the proposed big data processing.	3%	The Data Flow Diagram does not reflect the developed map-reduce algorithm used for big data processing.	The Data Flow Diagram somehow reflects the developed map-reduce algorithm used for big data processing	The Data Flow Diagram reflects the developed map-reduce algorithm used for big data processing.	The Data Flow Diagram highly reflects the developed map-reduce algorithm used for big data processing	**x/4 * 3 = /3**
8. Peer evaluation Rate each member (including yourself) on: • Initiative and responsibility • Communication and collaboration • Contribution to writing • Timeliness and reliability	2%	Poor	Satisfactory	Good	Excellent	**x/4 * 2 = /2**
PART 2: PROJECT PRESENTATION (10%)						
9. Project Presentation	10%	Demonstrate poor ability to present clearly with confidence and appropriate to the level of the listener/ presentation content does not reflect the given assignment.	Demonstrate moderate ability to present clearly with confidence and appropriate to the level of the listener/ somehow the presentation content reflects the given assignment.	Demonstrate adequate ability to present clearly with confidence and appropriate to the level of the listener/ the presentation content reflects the given assignment.	Demonstrate outstanding ability to present clearly with confidence and appropriate to the level of the listener/ the presentation content very much reflects the given assignment.	**x/4 * 10 = /10**
Total	**(40%)**					**/40**

END